

2023 春概率统计标准答案和评分标准

一、 1)B, 2)A, 3)A, 4)A, 5)C

二、

1)0.4; 2)0.271; 3)0; 4)0.1587; 5)2.2; 6)-14.8; 7)0.5; 8)4.2

$$9)F(x) = \begin{cases} 0 & x < -1 \\ 0.5 & -1 \leq x < 2 \\ 0.8 & 2 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

$$10)f_Y(y) = \begin{cases} 1.5y^2 - y + 1 & 0 < y < 1 \\ 0 & \text{其它} \end{cases}$$

11)2; 12)[24.175, 25.825]; 13)0.4

三、 1)

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1 \dots\dots\dots 1'$$

$$\int_0^1 \int_0^{1-y} (x + ay) dx dy = 1 \dots\dots\dots 2'$$

$$a = 5 \dots\dots\dots 2'$$

$$2)f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy \dots\dots\dots 1'$$

$$\text{当 } x < 0, \text{ or } x > 1 \text{ 时, } f_X(x) = 0; \dots\dots\dots 1'$$

$$\text{当 } 0 < x < 1 \text{ 时,}$$

$$f_X(x) = \int_0^{1-x} (x + 5y) dy = 1.5x^2 - 4x + 2.5 \dots\dots\dots 3'$$

$$3)f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} \dots\dots\dots 1'$$

$$= \begin{cases} \frac{x+5y}{1.5x^2-4x+2.5} & 0 < y < 1-x \\ 0 & \text{其他} \end{cases}, (0 < x < 1) \dots\dots\dots 1'$$

$$f_{Y|X}(y|0.5) = \begin{cases} \frac{4+40y}{7} & 0 < y < 0.5 \\ 0 & \text{其他} \end{cases}, \dots\dots\dots 1$$

$$P(0.2 < Y < 0.75|X = 0.5) = \int_{0.2}^{0.5} \frac{4+40y}{7} dy \dots\dots\dots 1'$$

$$= \frac{27}{35} \approx 0.771 \dots\dots\dots 1'$$

四、

B – 花费时间不超过 50 分钟

A – 采取自驾出行；

$$P(A) = 0.6, P(\bar{A}) = 0.4; \dots\dots\dots 1'$$

$$P(B|A) = \frac{3}{8}; P(B|\bar{A}) = \frac{1}{2} \dots\dots\dots 1'$$

$$(1) P(B) = P(A)P(B|A) + P(\bar{A})P(B|\bar{A}) \dots\dots\dots 2'$$

$$= 0.425 = 17/40 \dots\dots\dots 2'$$

$$(2) P(A|B) = \frac{P(A)P(B|A)}{P(B)} \dots\dots\dots 2'$$

$$= \frac{0.6 \times \frac{3}{8}}{0.575} = \frac{9}{17} \approx 0.529 \dots\dots\dots 2'$$

五

$$f_X(x) = \begin{cases} e^{-x} & x \geq 0 \\ 0 & x < 0 \end{cases}; f_Y(y) = \begin{cases} 2e^{-2y} & y \geq 0 \\ 0 & y < 0 \end{cases} \dots\dots\dots 1'$$

$$f(x, y) = f_X(x)f_Y(y) \dots\dots\dots 2'$$

$$F(z) = P(Z \leq z) = P(X + 2Y \leq z) \dots\dots\dots 1'$$

$$\text{当 } z < 0 \text{ 时, } F(z) = 0; \dots\dots\dots 1'$$

$$\text{当 } z \geq 0 \text{ 时, } F(z) = \int_0^z \int_0^{z-2y} 2e^{-x-2y} dx dy = 1 - e^{-z} - ze^{-z} \dots\dots\dots 3'$$

$$f(z) = [F(z)]' \dots\dots\dots 1'$$

$$= \begin{cases} ze^{-z} & z \geq 0 \\ 0 & \text{其他} \end{cases} \dots\dots\dots 1'$$

(也可以先算 $2Y$ 是 $e(1)$, 用卷积公式, 或者伽马分布的性质)

六、

设 100 件产品中非三级品个数为 X , 则

$$X \sim b(n, p), \text{ 其中, } n = 100; p = 0.9; \dots\dots\dots 2'$$

$$\mu = EX = np = 90; \sigma^2 = \text{var}(X) = np(1-p) = 9; \dots\dots\dots 2'$$

所求概率为

$$\begin{aligned} P(X > 92) &\approx 1 - \Phi\left(\frac{92 - \mu}{\sigma}\right) \\ &= 1 - \Phi\left(\frac{2}{3}\right) \dots\dots\dots 5' \end{aligned}$$

(也可以用 Y 表示三级品的个数, $Y \sim b(n, p)$, 其中, $n = 100; p = 0.1$, 计算 $P(Y \leq 8)$)

七、

1) 当 $x = 0$ 时: $f(0,1) < f(0,2), \hat{\theta} = 2 \dots \dots \dots 1'$

当 $x = 1$ 时: $f(1,1) < f(1,2), \hat{\theta} = 2 \dots \dots \dots 1'$

当 $x = 2$ 时: $f(2,1) > f(2,2), \hat{\theta} = 1 \dots \dots \dots 1'$

2) $\hat{\theta}$ 的分布.

当 $\theta = 1$ 时: $P(\hat{\theta} = 1) = f(2,1) = 0.5; P(\hat{\theta} = 2) = 0.5 \dots \dots \dots 2'$

$E\hat{\theta} = 1 \times 0.5 + 2 \times 0.5 = 1.5 \neq 1$ 。所以 $\hat{\theta}$ 不是 θ 的无偏估计量。.....2'

3) 当 $X = 1$ 时, $f(1,1) < f(1,2), \therefore \hat{\theta} = 2 \dots \dots \dots 3'$

八、

(1) 检验统计量: $T = \frac{\bar{X} - \mu_0}{S_n} \sqrt{n} \sim t(n-1) (H_0 \text{ 成立时})$

拒绝域: $D = \{T < -t_{\alpha}(n-1)\} \dots \dots \dots 3'$

其中, $n = 25, \mu_0 = 32, S_n = 4, \alpha = 0.05, t_{\alpha}(n-1) = 1.711;$

T 的观测值: $T = \frac{30 - 32}{4} \sqrt{25} = -2.5 \dots \dots \dots 1'$

$U = -2.5 < -1.711 \dots \dots \dots 1'$

所以, 拒绝原假设。.....1'

2) σ^2 的置信区间: $\left[\frac{(n-1)S_n^2}{\chi_{\frac{\alpha}{2}}^2(n-1)}, \frac{(n-1)S_n^2}{\chi_{1-\frac{\alpha}{2}}^2(n-1)} \right] \dots \dots \dots 2'$

$\alpha = 0.05; S_n^2 = 16, \chi_{\frac{\alpha}{2}}^2(n-1) = 39.36, \chi_{1-\frac{\alpha}{2}}^2(n-1) = 12.4; \dots \dots \dots 1'$

所求置信区间为: $[9.756, 30.968] \dots \dots \dots 1'$